Sleep-Related Respiratory Disruption is Associated with Altered Spindle Morphology and Poorer Attention in Children

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Introduction

- SDB affects ~1–5% of children; even mild forms (snoring, upper-airway resistance) are linked to neurocognitive impairment, behavioral dysregulation, and reduced academic performance [1].

- The **hypopnea index (HI)** captures partial airway obstruction that fragments sleep through intermittent hypoxemia and arousal [1] — disruption especially relevant to sleep microarchitecture.

- **Sleep spindles** are thalamocortical oscillations (10–16 Hz) of NREM sleep [2]. *Slow* spindles (~10–12 Hz) predominate frontally; *fast* spindles (~12–16 Hz) predominate centroparietally, reflecting distinct circuits and functions [3]. Fast spindles in particular engage hippocampal–cortical pathways [4].

- Spindle features (density, amplitude, duration, frequency) are **separable dimensions** of microarchitecture, not interchangeable markers.

- Most SDB–spindle work is in adults; the few pediatric studies used limited-montage polysomnography, leaving spindle *topography* uncharacterized [5].

- Childhood is a critical period for thalamocortical maturation, so it is unclear whether respiratory disruption causes a *global* loss of spindles or a *regionally selective* change in specific features.

Aims: (1) confirm canonical slow-anterior / fast-posterior organization; (2) test whether HI relates to regional spindle morphology; (3) determine whether HI-sensitive features predict attention.

Cohort Overview

	Value	n
Demographics		
Age (years)	7.7 (3.0, 11.0)	62
Boys / Girls	33 (53%) / 29 (47%)	62
Respiratory indices		
Hypopnea index (events/hr)	3.1 (0.1, 34.1)	62
Sleep metrics		
Spindle density (events/min)		
Slow (10–12 Hz)	0.6 (0.0, 2.0)	62
Fast (12–16 Hz)	0.4 (0.0, 1.9)	62
Spindle amplitude (μV)		
Slow (10–12 Hz)	55.3 (33.8, 94.0)	62
Fast (12–16 Hz)	54.8 (27.6, 79.8)	62
Spindle duration (s)		
Slow (10–12 Hz)	0.9 (0.7, 1.0)	62
Fast (12–16 Hz)	0.8 (0.6, 1.0)	62
Attentional performance (TOVA Q1)		
d-prime	4.6 (0.5, 8.5)	56
Omission errors	9.5 (0.0, 52.2)	56
Commission errors	3.2 (0.0, 29.5)	56
Mean reaction time (ms)	607.9 (353.0, 838.4)	56

Table 1. *Demographic characteristics, sleep spindle metrics, and attentional performance.*

Methods

- **Recording:** Overnight 256-channel HydroCel hdEEG (500 Hz) with concurrent clinical polysomnography; respiratory events scored to AASM criteria (HI = hypopneas/hour).

- **Attention:** Test of Variables of Attention (TOVA), a continuous performance test [6]; analyses focused on Quarter 1 (low-target-frequency vigilance). Outcomes: d-prime, omission errors, commission errors, reaction time.

- **Preprocessing (MATLAB / EEGLAB):** filtering, Zapline-plus line-noise removal, bad-channel rejection/interpolation, average reference; standardized 172-channel montage.

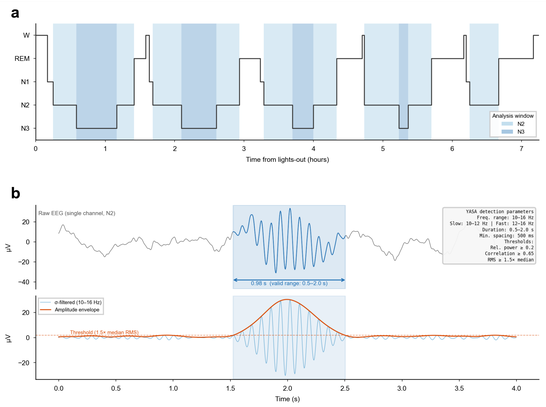


Figure 1. *Sleep architecture and spindle detection. Representative hypnogram with the N2+N3 analysis window, and the YASA detection procedure applied per channel.*

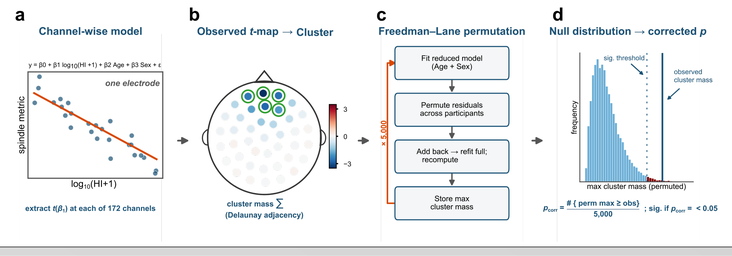


Figure 2. *Statistical pipeline. Per-channel regression of each spindle metric on hypopnea index (age- and sex-adjusted) yields a t-map; contiguous channels with $|t| > 2$ form clusters whose mass is tested against a Freedman–Lane permutation null (5,000 iterations) for multiple-comparison–corrected p-values.*

Results

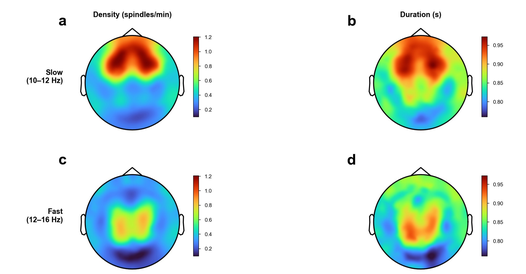


Figure 3. *Normative spindle topography. Group-averaged density and duration maps show the canonical slow-anterior / fast-centroparietal organization across the cohort.*

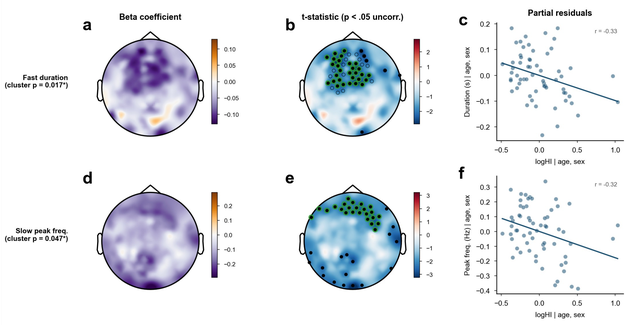


Figure 4. *Cluster-corrected HI effects. Two topographic effects survived permutation correction: shorter anterior fast spindle duration (corrected $p = 0.017$) and slower anterior slow spindle frequency (corrected $p = 0.047$) with rising HI.*

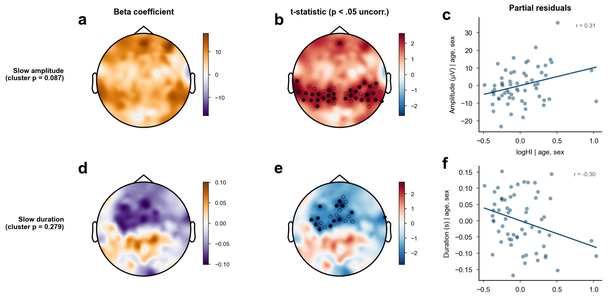


Figure 5. *Exploratory (uncorrected) effects. Posterior slow spindle amplitude increased and anterior slow spindle duration shortened with HI; neither survived cluster correction.*

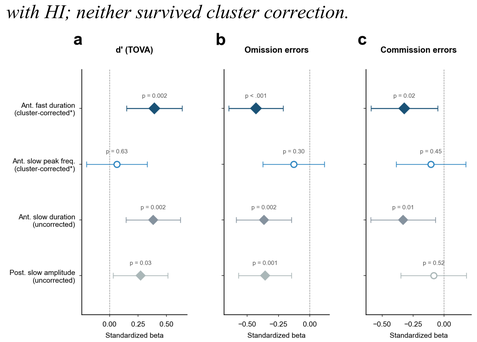


Figure 6. *Spindle features and attention (TOVA, N = 56). Standardized coefficients (age-, sex-, HI-adjusted) show anterior fast spindle duration as the most consistent predictor — higher d-prime, fewer omission and commission errors.*

Discussion

- The functionally relevant abnormality is a **loss of temporal stability in anterior fast spindles**, not a failure to generate spindles. Duration likely reflects the stability of thalamocortical oscillatory bouts [2].

- **Mechanism:** respiratory arousals may truncate spindle events before completion, narrowing the window for sleep-dependent plasticity; the fast-spindle, anterior topography fits their reliance on extended cortical/hippocampal networks [4].

- **Clinical context:** adenotonsillectomy reliably improves respiratory outcomes, yet cognitive benefit is inconsistent — the largest RCT found no neurocognitive improvement despite respiratory gains [8]. A spindle readout could explain why children with similar respiratory burden differ cognitively.

- **Limitations:** the cross-sectional design cannot establish whether respiratory disruption shortens spindles or shared maturational factors drive both, and the behavioral test was limited to TOVA Quarter 1 (sparse-target vigilance).

- **Next steps:** longitudinal pre/post-treatment hdEEG is the clearest test of directionality — if fast spindle duration normalizes with improved breathing and tracks attentional gains, it would strengthen the mechanistic interpretation; if not, the phenotype may instead mark a persistent neurodevelopmental vulnerability.

Source localization and spindle–slow-oscillation coupling could further localize the affected thalamocortical generators.

Conclusion

- Pediatric SDB may be understood not only as disturbed breathing during sleep, but as a **disruption of the developing sleep physiology that supports daytime cognitive control**.
- **Shortened anterior fast spindle duration** is a candidate physiological readout linking respiratory burden to attentional vulnerability.
- Localized spindle integrity may track neurocognitive risk **beyond conventional respiratory indices**, which describe the airway but not the brain.

References

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